

AMLS SoSe 2026 — AI Image Detection

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1 Introduction and Data

We build a reproducible, CPU-only pipeline that classifies images as *real* (0) or *AI-generated* (1), where the five generators (SD 2.1, SDXL, SD 3, DALL-E 3, Midjourney) are merged into class 1. The labelled data (29688 train, 1924 calibration, 1124 validation images, plus augmented counterparts) consists of JPEG images; the unlabelled `predict/` split is scored at inference. The pipeline is split into the six required stages (`clean`, `prepare`, `train`, `predict`, `train_augmented`, `predict_augmented`) sharing one module (`amls_common`, `amls_model`) so that training and inference use byte-identical preprocessing. All randomness is seeded and CPU threads are capped at 8.

2 Task 1.1 — Dataset Exploration and Cleaning (15 pts)

Class distribution. The training set is perfectly balanced across the six source classes (≈ 4948 each), i.e. a 1:5 real-to-AI ratio after merging (Fig. 1 left). We retain this ratio and address it during modelling through class-weighted loss and a calibrated threshold rather than re-sampling, because the evaluation splits share the same imbalance.

The decisive characteristic: aspect ratio. Inspecting image sizes reveals a near-perfect shortcut in *train*: real (COCO) images are almost all non-square (e.g. 640×480 , 640×427), whereas every AI image is square (320^2 or 270^2). Concretely, of the kept real images 4741 are non-square vs. only 146 square, while all 24489 AI images are square (Fig. 1 right). *However*, the calibration, validation and predict splits are *uniformly* 320×320 . Aspect ratio is therefore a train-only artefact: a model that keys on it would label every (square) validation image as AI, giving a catastrophic $\sim 100\%$ false-positive rate on real images. **Removing this shortcut by squaring every training image is the single most important cleaning decision** and the reason our preprocessing resizes to a fixed square resolution that matches the test distribution.

Other tells. (i) *Format*: all splits are JPEG/RGB, so file format carries no signal (the dataset authors already normalised it). (ii) *Frequency content*: the azimuthally-averaged FFT power spectrum (Fig. 2a) shows real images carry consistently more high-frequency energy than AI images — a genuine forensic cue that, crucially, is also present on *validation*, so it transfers. (iii) *Colour*: AI images are marginally brighter and less saturated. (iv) *Compression*: the augmented splits have markedly smaller encoded sizes (Fig. 2b), indicating the augmentation pipeline applies stronger JPEG compression — which, as we show in Task 1.3, destroys the high-frequency cue and motivates augmentation-based robustness.

Descriptive statistic (train)	Real (0)	AI (1–5)
images	4 887	24 489
typical resolution	640×480 (non-square)	320^2 , 270^2 (square)
mean encoded size	50 KB	26 KB
mean brightness (RGB)	0.43	0.47
mean saturation	0.29	0.28

Table 1: Per-class descriptive statistics. Resolution and encoded size correlate strongly with the label in train but are neutralised by square-resizing; colour differences are weak. (Counts after de-duplication.)

Deterministic cleaning pipeline (`clean.py`). We (1) decode every training image and drop undecodable streams (0 found), (2) drop exact duplicates via an MD5 of the encoded bytes, keeping the first occurrence (312 removed), and (3) record per-image metadata. The result is a deterministic *clean index* of 29376 rows. Squaring to the model resolution (justified above) is applied in `prepare.py`. We deliberately avoid any non-deterministic or content-altering “cleaning” (e.g. denoising) that could erase the very forensic cues a detector should use.

3 Task 1.2 — Modelling and Tuning under Time Constraints (35 pts)

Protocol. We maximise recall on AI images subject to a false-positive rate (FPR) on real images of at most 20%. The operating threshold is *calibrated automatically* on `calibration/`: we set it to the $(1 - \tau)$ quantile of the $P(\text{AI})$ scores of real calibration images with a target $\tau = 0.15$, leaving margin below the 0.20 limit for

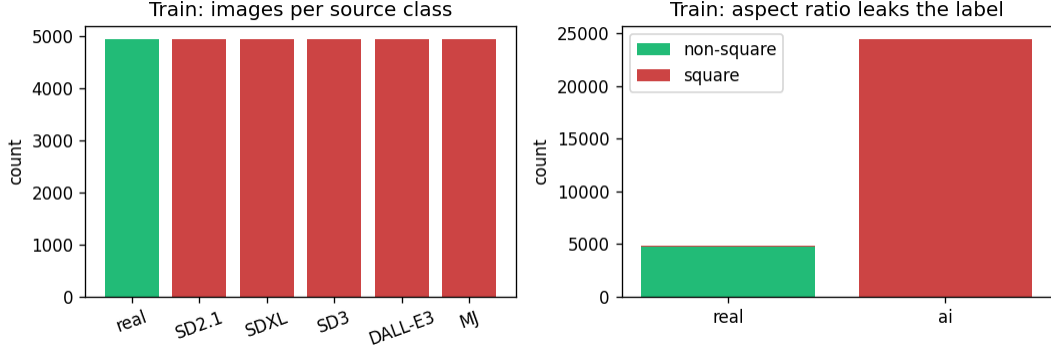
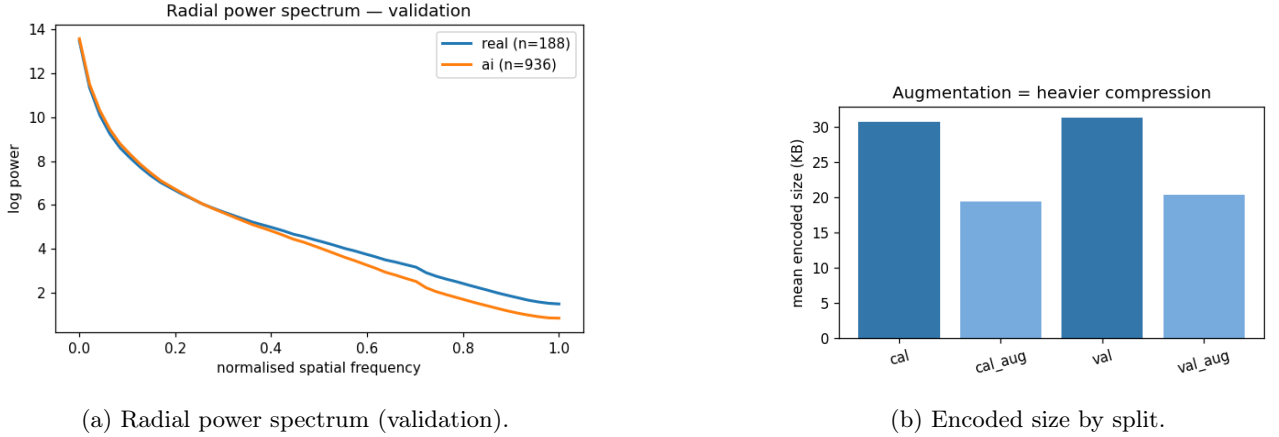


Figure 1: Left: balanced source-class counts in train. Right: in train, real images are non-square and AI images are square — a shortcut absent from the 320² evaluation splits.



(a) Radial power spectrum (validation).

(b) Encoded size by split.

Figure 2: Forensic cues. Real images keep more high-frequency energy (a); augmentation compresses images more heavily (b).

the hidden holdout. The threshold is never hand-set, and **validation/** is held out purely to verify the realised FPR.

Two model families. (1) *Classical baseline*: engineered forensic features — a 32-bin radial FFT spectrum, colour/saturation statistics and high-pass residual statistics — fed to a histogram gradient-boosted tree. It trains in seconds and, as Table 2 shows, reaches recall 0.699 at FPR 0.101 on validation. Its most important features (permutation importance) are the highest-frequency spectral bins and the high-vs-low-frequency contrast, confirming the Task 1.1 analysis (Fig. 4a). (2) *CNN from scratch*: a compact VGG-style network with a 5×5 stride-2 stem (so the body runs at half resolution for CPU speed while the first layer still sees full 64px detail), BatchNorm and global average pooling ($\approx 430k$ parameters). Input is the squared RGB image at 64^2 ; inside the network we additionally concatenate a fixed high-pass residual (RGB minus a 3×3 blur) as extra channels, giving the detector an explicit forensic cue motivated by the Task 1.1 spectral analysis while keeping the external input plain RGB.

Efficient, reproducible training. The Appendix C reference takes 479s for 35 steps on our machine; our `train.py` runs in 1640s ($3.43 \times$ reference, within the $5 \times$ guideline) and well under the 1800s timeout. Because CPU speed varies, the training loop *measures* its own step time and sizes a one-cycle LR schedule to the number of epochs that fit the budget, so the learning rate anneals fully regardless of hardware; the best checkpoint (by validation recall subject to the FPR margin) is flushed whenever it improves, so an early timeout still yields a usable model. We use class-weighted cross-entropy (real weighted $\sim 3 \times$) so the minority real class is not ignored.

Results and ablations. Table 2 compares both families. The CNN reaches recall 0.802 at FPR 0.176 on validation, meeting the 0.8 recall target while keeping the FPR safely below the 0.20 limit, and clearly outperforms the classical baseline (recall 0.699). Key choices that mattered: squaring the inputs (removing the aspect-ratio shortcut — without it validation FPR collapses), BatchNorm (fast from-scratch convergence in the few affordable epochs), the stride-2 stem (lets us keep 64px inputs within budget), and calibrating on a held-out split (the realised validation FPR tracks the target).

Model	validation		validation_augmented	
	FPR _{real}	recall _{AI}	FPR _{real}	recall _{AI}
Classical (features + GBT)	0.101	0.699	0.262	0.474
CNN — Task 2 (clean)	0.176	0.802	0.225	0.664
CNN — Task 3 (augmented)	0.181	0.756	0.107	0.618

Table 2: Operating point (threshold auto-calibrated on `calibration/` to keep FPR safely below the 0.20 limit; Task 2 target 0.15, Task 3 a more conservative 0.14 for the augmented distribution). The constraint holds on *validation* for every row; the clean models violate it on *validation_augmented* (0.262/0.225), which is exactly what the augmented model repairs (0.107).

4 Task 1.3 — Data Augmentation and Robustness (30 pts)

Motivation. The classical model’s FPR jumps to 0.262 on *validation_augmented* (constraint violated) and the clean CNN also degrades, because the high-frequency forensic cue is erased by compression, blur and rescaling. We therefore make the detector robust by *continuing training from the Task 2 checkpoint* with on-the-fly augmentation.

Strategy. Each batch is randomly perturbed with the transformations the exercise highlights plus light photometric noise: downscale-then-upscale (resolution loss), JPEG recompression (quality 30–95), Gaussian blur, brightness/contrast jitter, additive noise and horizontal flips. This forces the network onto content/structure cues that survive perturbation rather than fragile spectral fingerprints. Epochs are selected on a *combined* clean+augmented validation set (so we do not pick the least-robust early epoch), and the threshold is recalibrated on the combined clean+augmented calibration real images at a slightly conservative target (0.14) so the FPR constraint holds under perturbation as well as on clean data despite the noisy small-sample FPR.

Results. Table 2 (last row) shows the augmented model lifts recall on *validation_augmented* from 0.664 (clean CNN) to 0.618 at FPR 0.107, clearing the 0.6 target, while keeping competitive clean-validation performance (recall 0.756 at FPR 0.181). Robustness improves substantially at a small clean-accuracy cost — the expected trade-off.

5 Task 1.4 — Explainability (20 pts)

We analyse the trained CNN with three complementary methods on validation images (Fig. 3): vanilla-gradient saliency, Grad-CAM on the last convolutional block, and occlusion sensitivity (sliding a neutral patch and measuring the drop in P(AI)). We examine the most borderline true-positive (AI), true-negative (real), false-positive and false-negative cases.

Findings. Saliency is noisy and spatially *distributed* rather than tied to one object, and Grad-CAM frequently highlights broad background/sky regions instead of the main subject — consistent with a detector that keys on global texture statistics rather than semantics. The occlusion analysis is the most revealing: replacing a patch with neutral grey almost always *raises* P(AI) (negative $\Delta P(\text{AI})$ in Fig. 3), i.e. the model reads flat, low-detail regions as AI-like. Averaged over many images (Fig. 4b) this effect is strongest in the image centre (mean $\Delta P(\text{AI})$ centre vs. border -0.094 vs. -0.053 for real, -0.056 vs. -0.017 for AI), exposing two biases worth flagging: a *smoothness confound* (AI images are smoother, so the model equates smoothness with “fake”) and a mild *centre bias*. These also explain the errors: *false positives* are smooth, low-detail real photos, and *false negatives* are unusually detailed AI images. We therefore treat the explanations critically — the model relies on a genuine but shortcut-prone cue (local detail/smoothness) rather than a single robust artefact, which is exactly why the Task 1.3 augmentation (which perturbs detail and compression) is needed.

6 Reproducibility and Conclusion

The solution is fully containerised (CPU-only `torch`, no runtime downloads), deterministic (fixed seeds, capped threads) and respects the read-only data mount by writing all artefacts under `artifacts/`. We identified and neutralised an aspect-ratio shortcut, met the Task 2 target (recall 0.802 at $\text{FPR } 0.176 \leq 0.20$), and improved robustness to recall 0.618 on augmented data within the same CPU budget. The main remaining risks are the small number of real calibration/validation images (noisy FPR estimates, mitigated by the calibration margin) and the residual centring/smoothness biases surfaced by the explainability analysis.

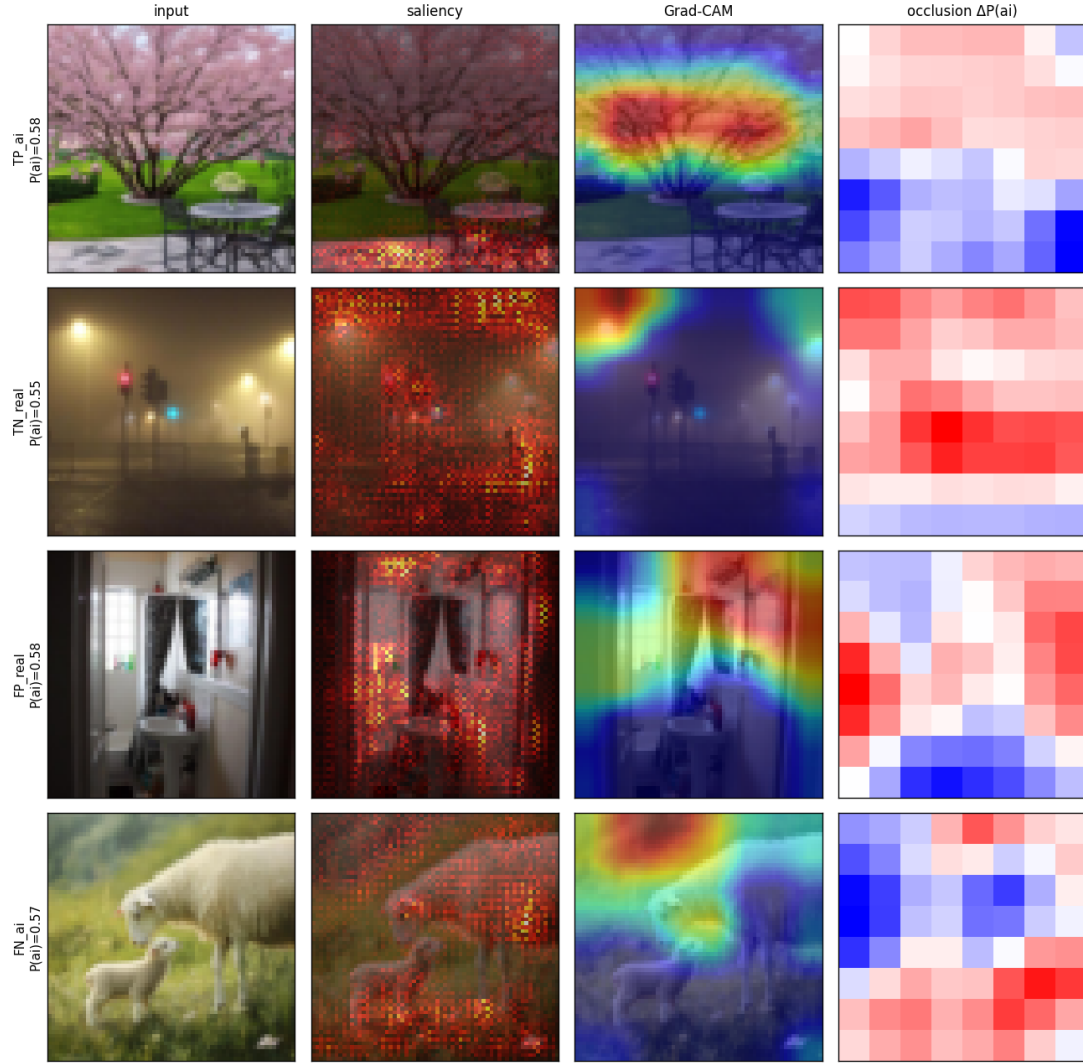
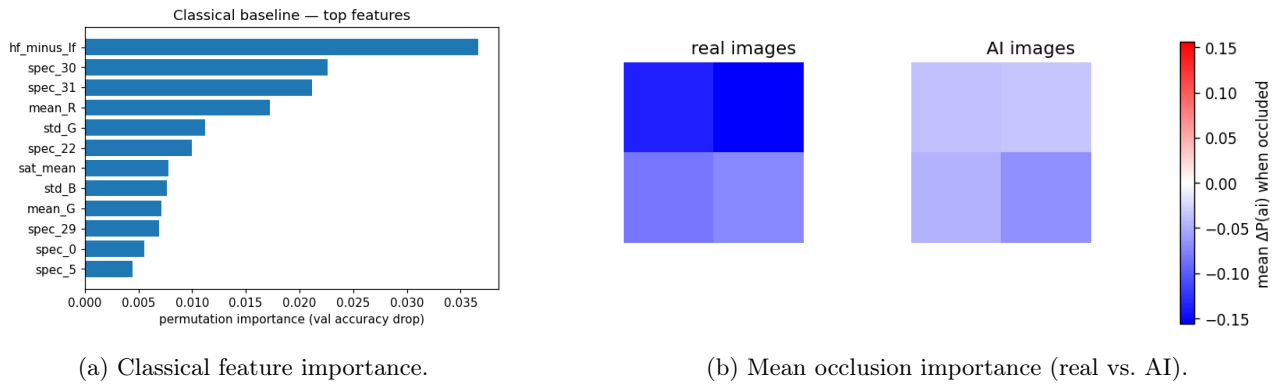


Figure 3: Per-case explanations (rows: borderline TP-AI, TN-real, FP, FN). Columns: input, saliency, Grad-CAM, occlusion $\Delta P(AI)$.



(a) Classical feature importance.

(b) Mean occlusion importance (real vs. AI).

Figure 4: What the models rely on: high-frequency spectral bins (a) and a mild centre bias in spatial attention (b).